Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.2 Area of Compound Shapes

Area of Compound Shapes

STARTER · 5 MIN

Date: _____ **Title: Area of Compound Shapes**

LEARNING OBJECTIVES

- Calculate the area of compound shapes

STARTER *Answer in your book. No calculators.*

1. Area of a rectangle 8 mm by 6 mm

2. Work out $\frac{1}{2} \times 8 \times 5$

3. Work out 140 − 53

4. Work out 12 − 5

Starter — Answers

STARTER

Tick or correct in green pen

1. **48 mm²**
2. **20**
3. **87**
4. **7**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify the simple shapes inside a compound shape	☐
SILVER	I can apply compound area to real-life problems	☐
GOLD	I can use the correct formulas and combine the areas	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Area of Compound Shapes

INTRODUCTION · 3 MIN

A compound shape is made from simple shapes: split it, then add or subtract.

Split

Draw a line to make simple shapes.

rectangles and triangles

Missing lengths

Use the lengths you know.

10 − 4 = 6 cm

Add

Add the areas of the parts.

48 + 20 = 68 mm^2

Subtract

Whole shape minus the hole.

140 − 24 = 116 cm^2

Examples 1 and 2 — Shapes Made of Rectangles

EXAMPLES · 5 MIN

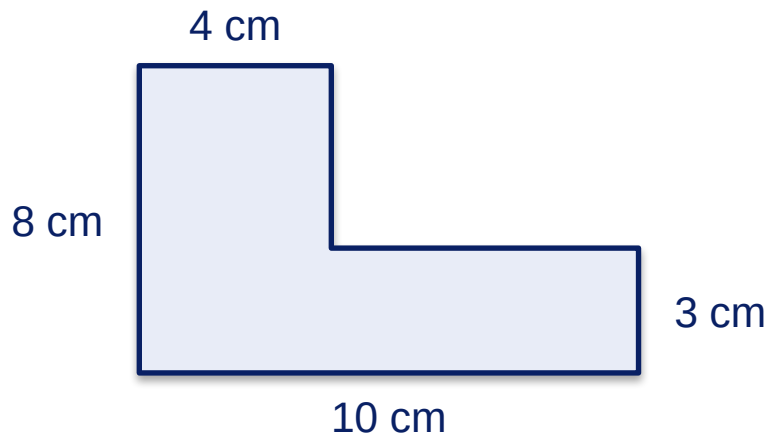
BRONZE

Watch and copy into your book

Hint Split into rectangles Find missing lengths

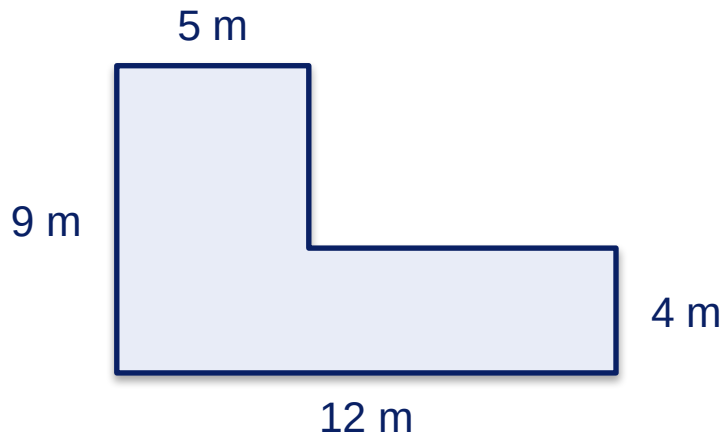
Example 1

Find the area.



Example 2

Find the area.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Split into two rectangles

$A = 4 \times 8 + 6 \times 3$

$= 32 + 18$

$\mathbf{= 50\ cm^2}$

Tip: the bottom part is 10 − 4 = 6 cm long.

Example 2

Split into two rectangles

$A = 5 \times 9 + 7 \times 4$

$= 45 + 28$

$\mathbf{= 73\ m^2}$

Tip: 12 − 5 = 7 m.

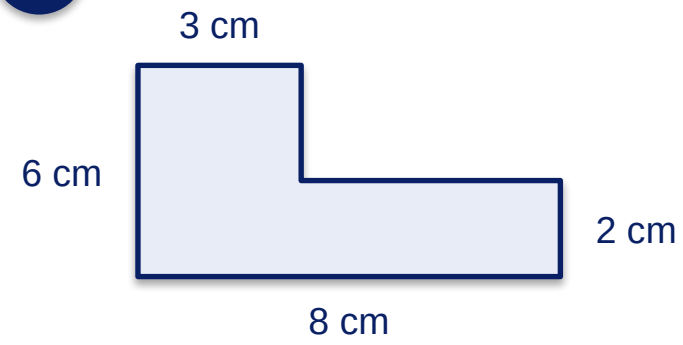
Task 1

PRACTICE · 9 MIN

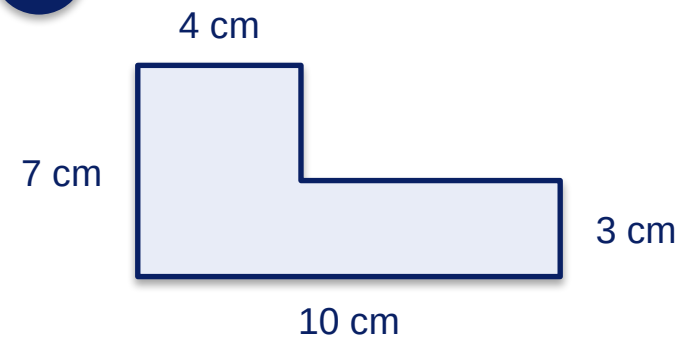
BRONZE *Find the area*

Hint Split into rectangles Find missing lengths

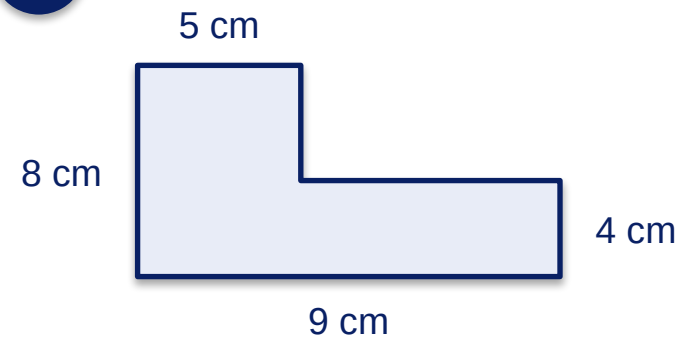
1 Area



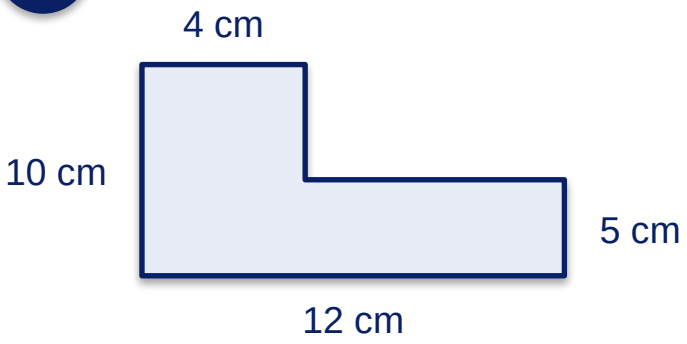
2 Area



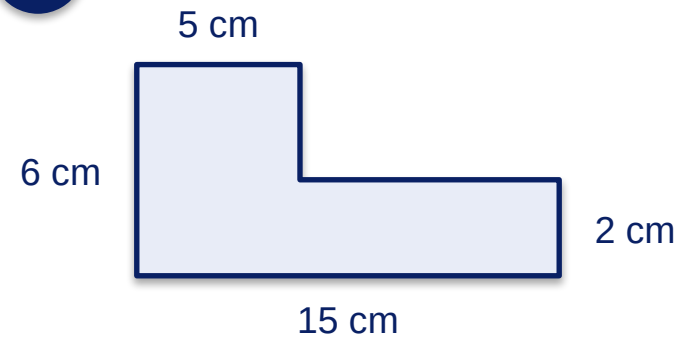
3 Area



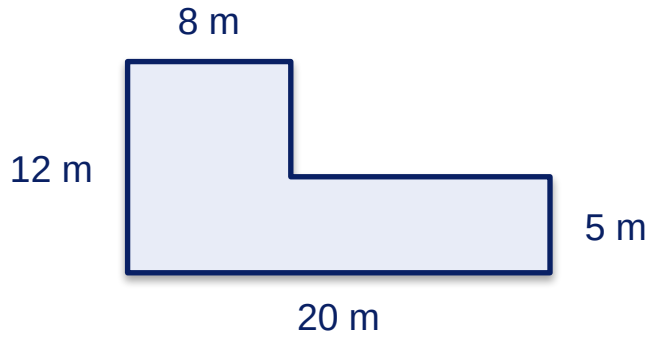
4 Area



5 Area



★ Area



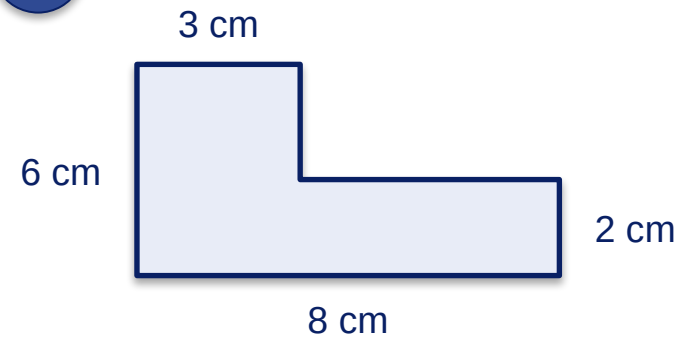
Task 1 — Answers

CORRECTIONS

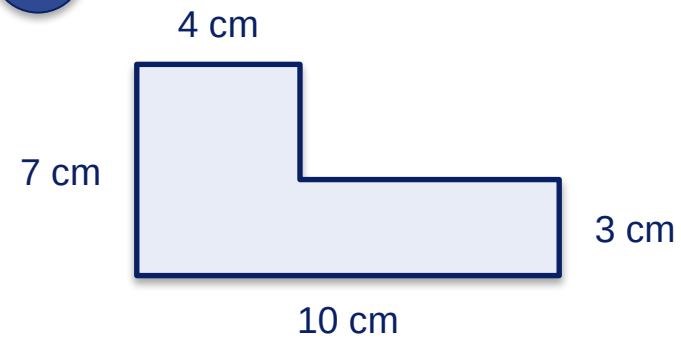
BRONZE

Tick or correct in green pen

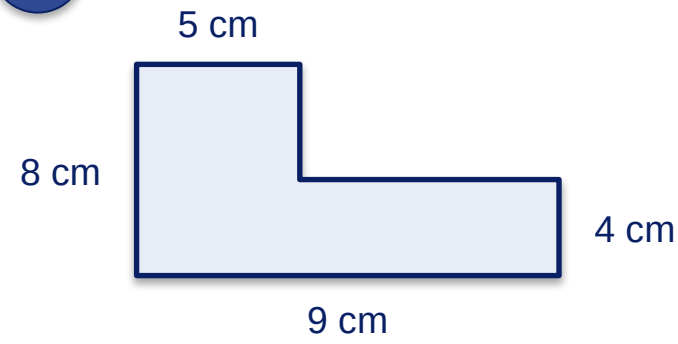
1 **28 cm²**



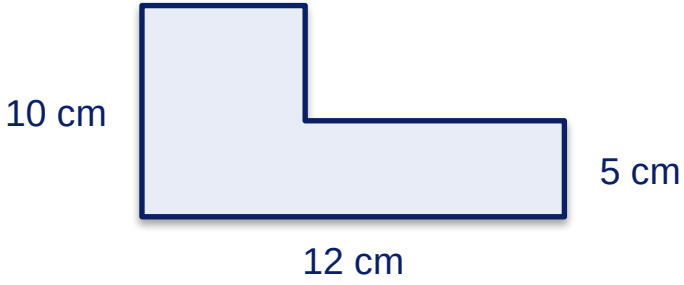
2 **46 cm²**



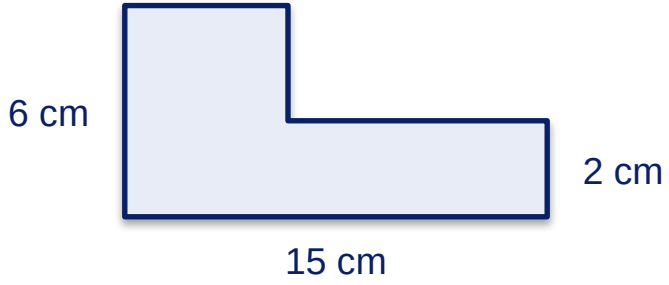
3 **56 cm²**



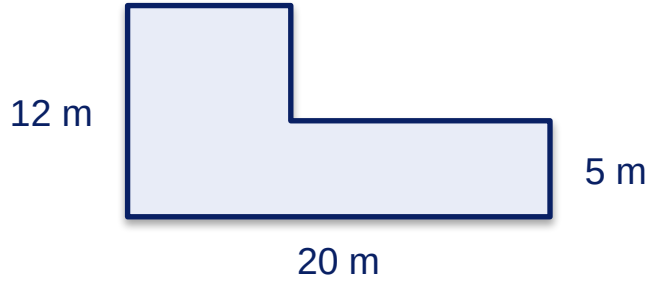
4 **80 cm²**



5 **50 cm²**



★ **156 m²**



Examples 3 and 4 — Rectangles and Triangles

EXAMPLES · 5 MIN

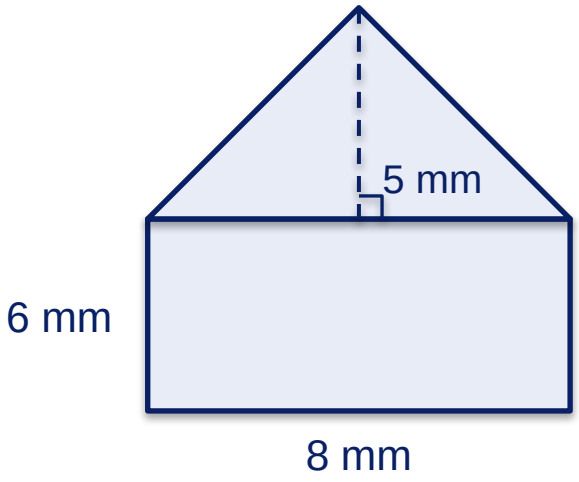
SILVER

Watch and copy into your book

Hint Add the parts Subtract a hole

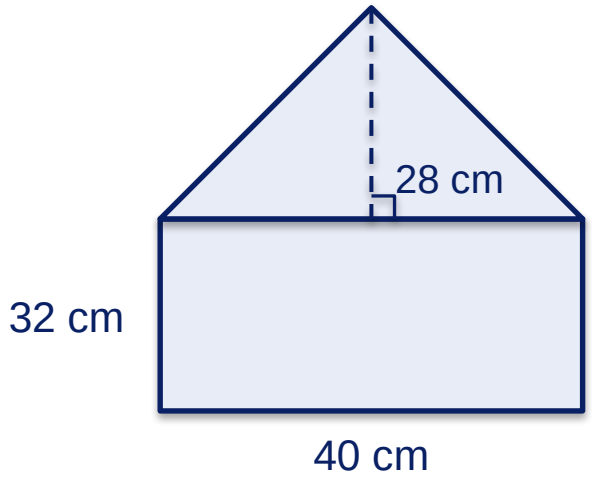
Example 3

Find the area.



Example 4

Find the area.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Rectangle + triangle

rectangle $= 8 \times 6 = 48$

triangle $= \frac{1}{2} \times 8 \times 5 = 20$

$\boldsymbol{A}$ **$= 68\ \text{mm}^2$**

Tip: the triangle's base is the 8 mm side.

Example 4

Rectangle + triangle

rectangle $= 40 \times 32 = 1280$

triangle $= \frac{1}{2} \times 40 \times 28 = 560$

$\boldsymbol{A}$ **$= 1840\ \text{cm}^2$**

Tip: label each part before you add.

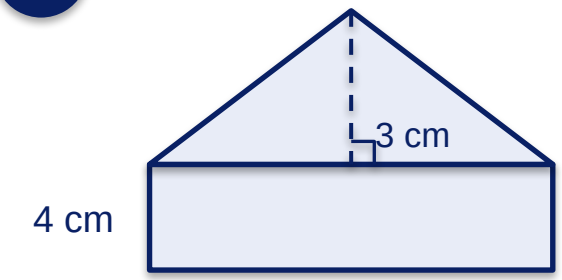
Task 2

PRACTICE · 9 MIN

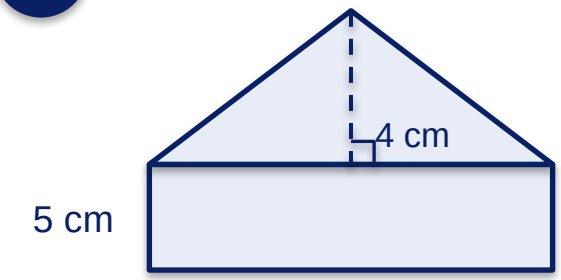
SILVER *Find the area*

Hint Add the parts Subtract a hole

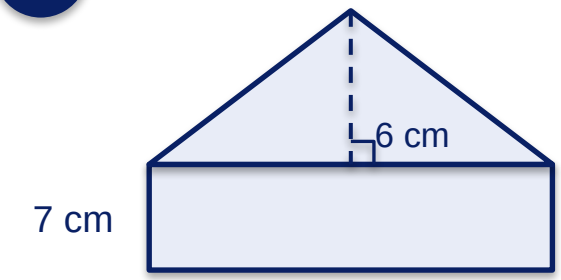
1 Area



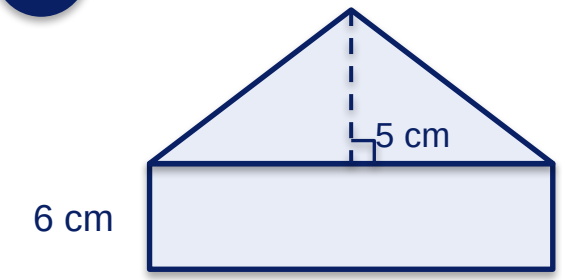
2 Area



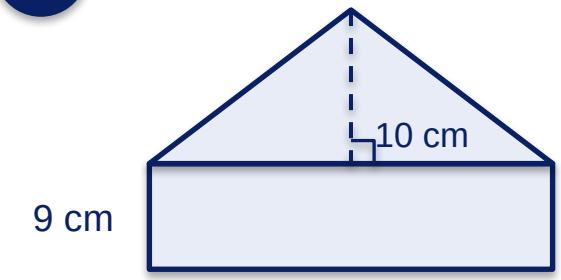
3 Area



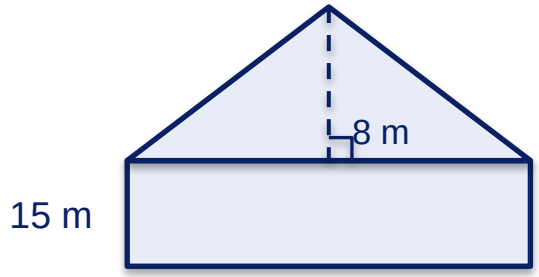
4 Area



5 Area



★ Area



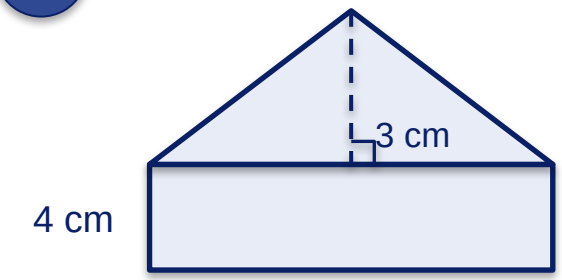
Task 2 — Answers

CORRECTIONS

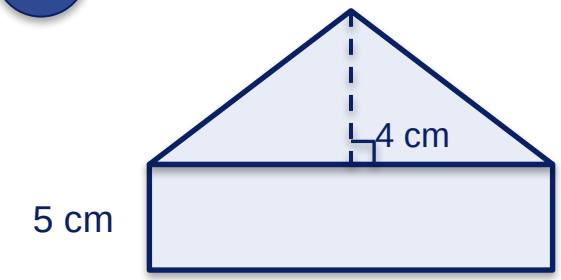
SILVER

Tick or correct in green pen

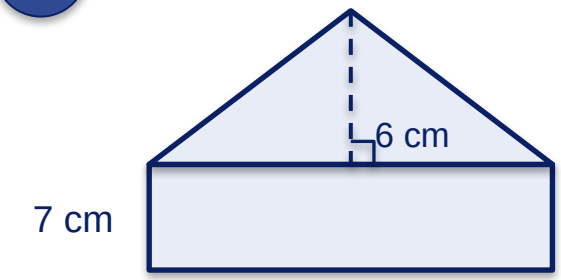
1 **33 cm²**



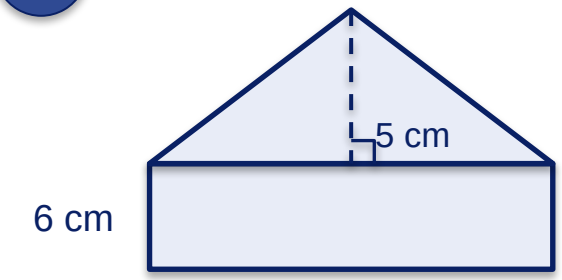
2 **70 cm²**



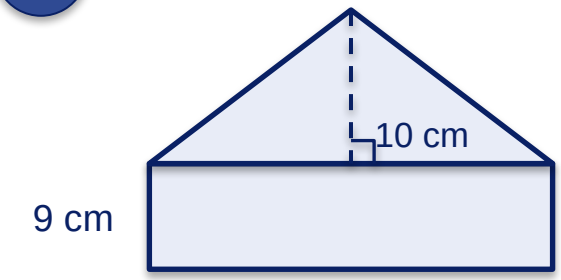
3 **80 cm²**



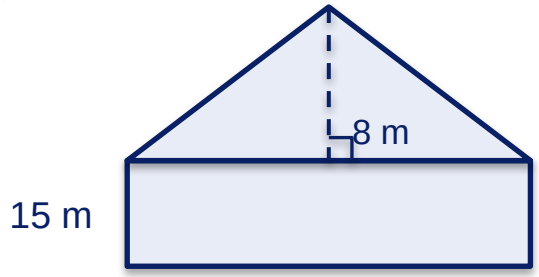
4 **102 cm²**



5 **196 cm²**



★ **380 m²**



Examples 5 and 6 — Shapes with Holes

EXAMPLES · 5 MIN

GOLD

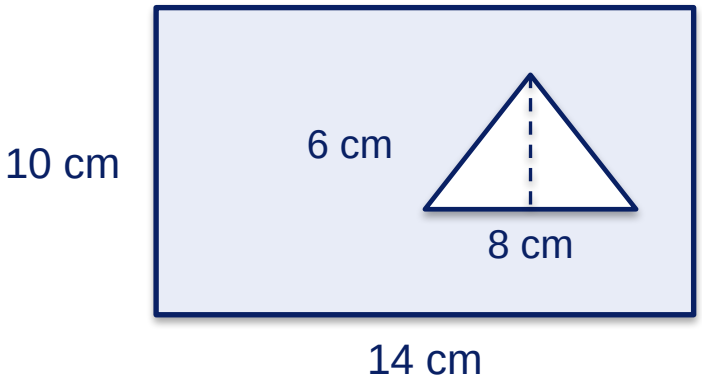
Exam-style: show all your working

Hint Area first, then cost Round tins up

Example 5

(3 marks)

A triangle is cut out of a plastic sheet. Find the area left.



Example 6

(4 marks)

A garden is 20 m by 12 m.

It has a square pond of side 4 m.

Seed costs £3 per m^2 of lawn.

Find the cost of the seed.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Whole rectangle − triangle

rectangle $= 14 \times 10 = 140$

triangle $= \frac{1}{2} \times 8 \times 6 = 24$

$\boldsymbol{A = 140 - 24 = 116\ \text{cm}^2}$

Tip: subtract the hole.

Example 6

Lawn = garden − pond

lawn $= 240 - 16 = 224\ \text{m}^2$

cost $= 224 \times 3$

$\boldsymbol{= £672}$

Tip: the pond does not need seed.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: area problems

Hint Area first, then cost Round tins up

1. Rectangle 12 cm by 8 cm with a 3 cm by 2 cm hole. Area left?

2. Square of side 10 cm with a square hole of side 4 cm. Area left?

3. Rectangle 9 m by 6 m minus a triangle: base 4 m, height 3 m

4. Floor 6 m by 5 m. Carpet costs £12 per m^2. Cost?

5. Wall 5 m by 3 m with a window 2 m by 1 m. Area to paint?

★ Lawn 15 m by 8 m, pond 3 m by 2 m. Seed £2 per m^2. Cost?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **90 cm²**
2. **84 cm²**
3. **48 m²**
4. **£360**
5. **13 m²**

★ **114 × 2 = £228**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE — True or false: a compound shape is made of several shapes

SILVER — True or false: you divide by 2 in every area formula

GOLD — True or false: a triangle's height is always a side

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **True**

SILVER **False: only triangles and trapezia**

GOLD **False: it is perpendicular to the base**

Exam Practice — Compound Shapes

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

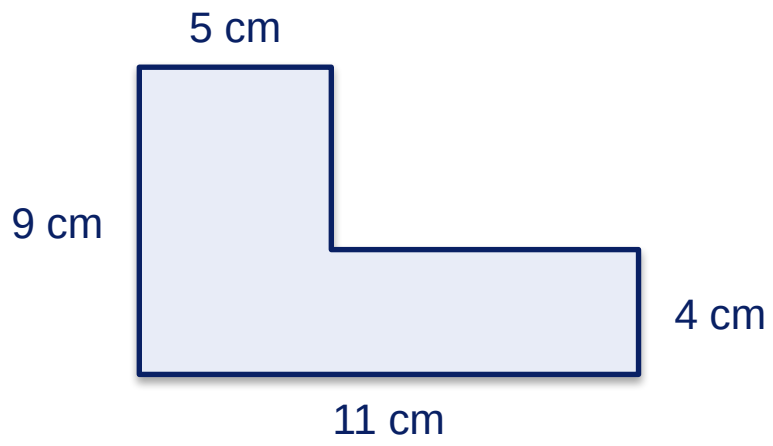
Hint Split into rectangles Find missing lengths

Question 1

(3 marks)

Work out the area of this shape.

Diagram NOT accurately drawn



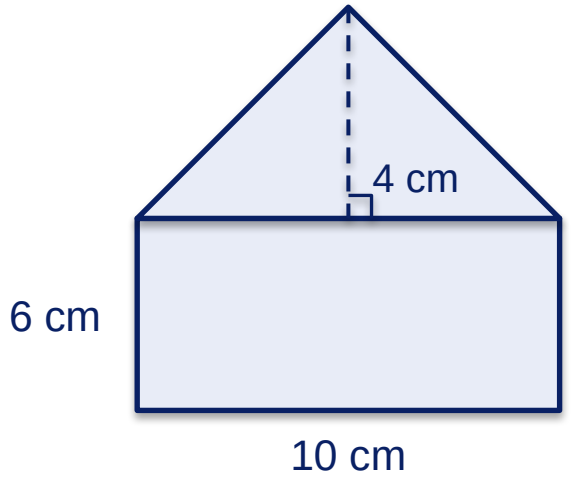
..... cm²

Question 2

(2 marks)

Work out the area of this shape.

Diagram NOT accurately drawn



..... cm²

Exam Practice — Holes and Triangles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Add the parts Subtract a hole

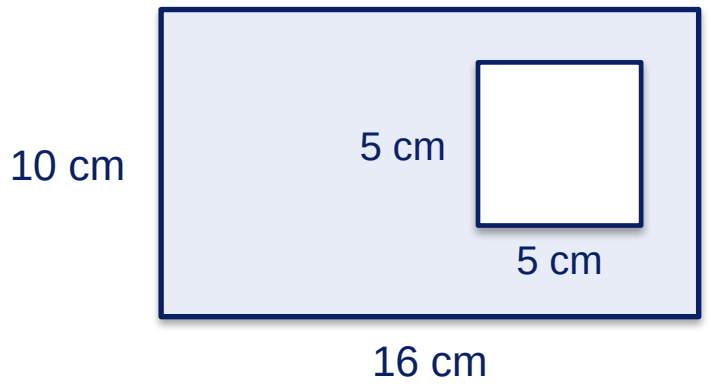
Question 3

(3 marks)

A metal plate has a square hole.

Work out the area of metal left.

Diagram NOT accurately drawn



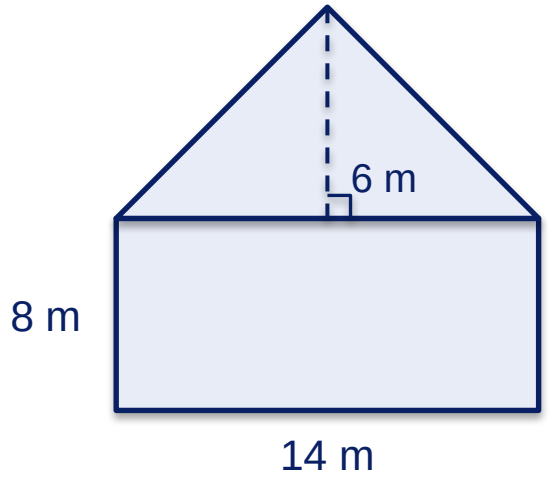
..... cm^2

Question 4

(3 marks)

Work out the area of this shape.

Diagram NOT accurately drawn



..... m^2

Exam Practice — Real-Life Area

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Area first, then cost Round tins up

Question 5

(4 marks)

A garden is a rectangle 18 m by 10 m.
It has a rectangular pond 4 m by 3 m.
Turf costs £5 per m².

Work out the cost of turf for the lawn.

..... £

Question 6

(4 marks)

A wall is 6 m by 2.5 m. It has a door 1 m by 2 m. One tin covers 5 m².

How many tins are needed to paint the wall?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$11 - 5 = 6$ cm — **B1**

$A = 5 \times 9 + 6 \times 4$ — **M1**

$= 69$ cm² — **A1**

Tip: find the missing length first.

Question 2

$A = 10 \times 6 + \frac{1}{2} \times 10 \times 4$ — **M1**

$= 80$ cm² — **A1**

Tip: 60 + 20 = 80.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

plate $= 16 \times 10 = 160$ — M1

hole $= 5 \times 5 = 25$ — M1

A $\mathbf{= 135\ cm^2}$ — A1

Tip: subtract the hole.

Question 4

rectangle $= 14 \times 8 = 112$ — M1

triangle $= \frac{1}{2} \times 14 \times 6 = 42$ — M1

A $\mathbf{= 154\ m^2}$ — A1

Tip: add the two parts.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

garden	= 18 × 10 = 180	M1
lawn	= 180 − 12 = 168 m²	A1
cost	= 168 × 5	M1
	= £840	A1

Tip: subtract the pond before the cost.

Question 6

wall	= 6 × 2.5 = 15	M1
paint	= 15 − 2 = 13 m²	A1
tins	= 13 ÷ 5 = 2.6	M1
	= 3 tins	A1

Tip: round up: 2 tins is not enough.